

SUMMARY TABLE: CORE MATHEMATICS GRADE 10

Sub-Strand 1.3: Quadratic Expressions and Equations — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 1.3: Quadratic Expressions and Equations
Driving Question	DRIVING QUESTION: "How can we use the mathematics of quadratic expressions and equations to predict, explain, and solve real problems — from the arc of a thrown ball on our school field to the dimensions of a farmer's shamba in Kericho?"

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Noticing the Curve — Introducing Quadratic Expressions Through the Shot Put Phenomenon	Learners observed that when the height and horizontal distance measurements from the shot put video are plotted on squared paper, the 7 data points form a smooth arch that cannot be fitted by a straight line — falsifying the linear model $y = mx + c$.	Learners learned that a quadratic expression (containing an x^2 term as its highest power) is needed to describe a curved relationship; that constant second differences in a data table signal a quadratic pattern; and that quadratic expressions can be formed by expanding products of two linear brackets.	Learners explained (and began to model) why the shot put's path curves — the x^2 term in the expression $h = -0.1x^2 + 1.2x + 1.5$ produces outputs that increase then decrease, matching the observed rise-and-fall of the ball, unlike the constantly-changing outputs of a linear equation.
Lesson 2 Forming Quadratic Expressions: From the Shot Put Arc to the Shamba	Learners watched the slow-motion shot put video still and worked with a height-distance data table from a throw at a Nairobi school sports day. They also examined a rectangular shamba diagram from Kericho, labelled with side lengths x and $(x + 4)$ metres, and computed first and second differences from the shot put height data.	Learners discovered that the area of the rectangular shamba produces an expression with an x^2 term when expanded — $x^2 + 4x$ — and that the second differences in the shot put height data are approximately constant, a signature of a quadratic relationship. They identified the standard form $ax^2 + bx + c$ and distinguished it from linear expressions by the presence of the x^2 term (degree 2).	The curved, parabolic path of the shot put cannot be modelled by a linear expression because the height data does not change by a constant amount per metre of horizontal distance. It requires a quadratic expression — one containing an x^2 term — because the second differences are constant. The same structure ($ax^2 + bx + c$) describes the area of a rectangular shamba when one or both dimensions depend on a variable, confirming that quadratic expressions arise naturally from both motion and area contexts in real Kenyan settings.
Lesson 3 Factorising Quadratic	Learners arranged algebra tiles into rectangles to find the factor-pairs of	Factorising a quadratic expression (both the $a=1$ and $a\neq 1$ cases using the product-sum	Learners annotated their cumulative parabola model with the factored form of

Expressions — Breaking the Curve's Code	quadratic expressions, and connected the factored form of $h = -x^2 + 6x$ to the shot put's launch and landing points on the sports-day video still.	and ac methods) reverses the expansion process; the two factors of a height expression reveal the exact horizontal distances where the ball is at ground level (height = 0).	the height expression and labelled the two zeros on the horizontal axis, and wrote a Model Update Statement explaining how factorisation connects to predicting the ball's landing point.
Lesson 4 Squaring the Shamba — Deriving Quadratic Identities Through Area	Learners physically cut and rearranged squared graph-paper pieces representing a Kericho farmer's square shamba, observing that a square of side $(a+b)$ always decomposes into exactly four regions — a^2 , ab , ab , and b^2 — no matter what values a and b take. They also observed that removing a square corner from a larger square produces an L-shaped area equal to $(a+b)(a-b)$, confirming that $a^2 - b^2$ factors into two binomials.	The three standard quadratic identities — $(a+b)^2 = a^2 + 2ab + b^2$, $(a-b)^2 = a^2 - 2ab + b^2$, and $(a+b)(a-b) = a^2 - b^2$ — are not arbitrary rules but geometric truths rooted in area. The middle term '2ab' in the perfect square identities represents the two rectangle strips that are always present when a square is partitioned, and its sign depends on whether the side is being extended or reduced. The difference of two squares identity provides a rapid factorisation shortcut for any expression of the form $a^2 - b^2$.	The shot put's height expression $-x^2 + 6x$ can be rewritten using the perfect square identity as $-(x-3)^2 + 9$, revealing that the ball reaches a maximum height of 9 metres at a horizontal distance of 3 metres — the turning point of the parabola. This shows that the identities derived from the shamba's area are not separate from the shot put phenomenon: they are tools for reading the hidden structure of the quadratic expression that describes the ball's arc, and they point directly toward the concept of solving quadratic equations (finding where $h = 0$) in the next lessons.
Lesson 5 Factorising Quadratic Expressions — Unlocking the Hidden Structure of the Curve	Learners completed a Factorisation Investigation Worksheet in groups of 4, discovering the product-sum rule by filling a structured table and applying it to factor six quadratic expressions — including a Kericho tea-shamba area context. They then verified each factorisation by re-expanding using the FOIL method from Lesson 4. Groups recorded findings on manila sheets for gallery sharing.	Learners learned that every quadratic expression of the form $x^2 + bx + c$ can be written as a product of two brackets $(x + p)(x + q)$, where $p \times q = c$ and $p + q = b$. They extended this to $ax^2 + bx + c$ ($a > 1$) using the split-middle-term and grouping method. Crucially, they discovered that factorising the shot put height model $-x^2 + 6x$ into $-x(x - 6)$ immediately reveals the two horizontal distances where height = 0 — $x = 0$ (launch) and $x = 6$ m (landing) — giving algebraic predictive power over the physical event.	Learners explained that factorisation is the reverse of expansion: it 'un-wraps' a quadratic expression to reveal its hidden bracket structure. They connected this to the anchoring phenomenon by annotating their cumulative parabola models with the factorised form and the two x-intercepts, and wrote 'Before/After' revision notes articulating how factorisation answers the question of where the shot put lands. The Driving Question Board was updated with the insight: 'The factorised form tells us where the ball hits the ground (the zeros/roots of the expression).'
Lesson 6 Expanding Products of Two Linear Brackets: Grid/Box Method and FOIL in Kenyan Contexts	Learners observed the teacher work through three fully annotated examples of bracket expansion using both the grid/box method and FOIL in parallel, with Kenyan contexts (Nakuru school garden, Kongowea fish seller, shot-put parabola). They observed that $(-x + 6)(x - 1)$ expands to $-x^2 + 7x - 6$ — the algebraic rule for the shot-put's curved path.	Learners learned that the product of two linear expressions $(ax + b)(cx + d)$ always produces a quadratic expression; that the grid/box method organises all four term-products systematically; that FOIL (First, Outer, Inner, Last) provides an equivalent structured procedure; and that like terms must be collected carefully — especially when negative signs are involved — to arrive at the standard form $ax^2 + bx + c$.	Learners explained and corrected common expansion errors (missing middle term, sign errors with negatives, incorrect collection of like terms) through structured pair practice and Error Trap analysis. They updated the class Driving Question Board and cumulative model to show that the shot-put's parabolic height rule $h = (-x + 6)(x - 1)$ is mathematically equivalent to $h = -x^2 + 7x - 6$, bringing the class closer to being able to predict where the ball lands.

Lesson 7 Forming and Solving Quadratic Equations in Real-Life Situations	Learners observed that setting $h = 0$ in the shot put equation $h = -x^2 + 6x$ produces a quadratic equation $x^2 - 6x = 0$, and that a real-world area constraint on a Kericho shamba similarly produces $x^2 + 3x - 40 = 0$ — demonstrating that the same algebraic structure emerges from very different physical situations.	Learners learned to form quadratic equations in standard form from real-life word problems, to solve them by factorisation (common factor and trinomial) and by the quadratic formula, and to interpret both roots in context — rejecting non-physical solutions such as negative lengths. They established that $x = 6$ m is the landing distance of the shot put and that the shamba dimensions are $5 \text{ m} \times 8 \text{ m}$.	Learners explained that the parabolic path of the shot put can be decoded algebraically: by setting the height equation equal to zero and solving, we predict exactly where the ball lands before it lands. The quadratic formula provides a universal method when factorisation is not obvious. A solution that is mathematically valid but physically impossible (negative dimension) must be rejected by interpreting the root in real-world context.
Lesson 8 Final Explanation: Reversing the Expansion — Factorisation, Difference of Two Squares, and the Full Picture of the Parabolic Path	Learners observed that the area expression $x^2 + 7x + 10$ from the tiled courtyard — which they had previously generated by expanding $(x+2)(x+5)$ — could be reversed: starting from the expanded form and recovering the two side-length factors $(x+2)$ and $(x+5)$ by finding a pair of numbers that multiply to 10 and add to 7. Learners also observed that the Kericho shamba's area expression $(x+9)(x-9)$ produced $x^2 - 81$, and that this 'difference of two squares' structure allowed instant factorisation using the identity $(a+b)(a-b) = a^2 - b^2$. Finally, learners observed that the shot-put height equation $h = -x^2 + 6x$, when set equal to zero, could be factorised as $-x(x-6) = 0$, yielding $x = 0$ and $x = 6$ — the exact horizontal distances at which the ball is at ground level.	Learners learned that factorisation is the reverse of expansion: every quadratic expression that arose from an area context was originally a product of two factors, and algebraic reasoning can recover those factors. They learned three factorisation methods: (1) extracting a common factor, (2) factorising a trinomial $x^2 + bx + c$ by finding two numbers that multiply to c and sum to b, and (3) applying the difference of two squares identity $a^2 - b^2 = (a+b)(a-b)$ as a special, efficient case. They learned that setting a factorised quadratic equation equal to zero and applying the zero-product property yields the roots — the values of the unknown that solve the equation — and that these roots carry real-world meaning: landing distances, shamba dimensions, courtyard side lengths.	The parabolic path of the shot put thrown on sports day in Nairobi is described by a quadratic equation. Quadratic expressions arise naturally from area — whenever two variable lengths are multiplied, the result is a quadratic. Factorisation reverses this process, breaking a quadratic expression back into the two lengths that produced it. When the height equation of the shot put is set to zero and factorised, the roots reveal exactly where the ball meets the ground. The difference of two squares identity is a powerful special case that arises when two lengths differ only in sign — as in the Kericho shamba with sides $(x+9)$ and $(x-9)$. Across all these contexts — a tiled courtyard, a Rift Valley shamba, and an athlete's throw — the same algebraic structure (the quadratic) and the same algebraic tool (factorisation) explain and predict real outcomes. The driving question is fully answered: quadratic expressions and equations are the mathematical language of curved, symmetric, real-world phenomena.

ATTRIBUTION AND LICENSING

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